Hypothesis testing with multiple regression

Hypothesis testing with a categorical predictor

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

We want to know whether there is a relationship between species and bill length.

> Can I translate this into null and alternative hypotheses about one or more model parameters?

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$H_0 : \beta_1 = \beta_2 = 0$

$H_A :$ at least one of $\beta_1, \beta_2 \neq 0$

Full and reduced models

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

$H_0 : \beta_1 = \beta_2 = 0$

$H_A : \text{at least one of } \beta_1, \beta_2 \neq 0$

Reduced model (H_0):

$$\text{bill length} = \beta_0 + \varepsilon$$

Full model (H_A):

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

> What test did we use to test these hypotheses?

F test

Variability:

$$\sum_{i=1}^{n}(y_i - \overline{y})^2 = \sum_{i=1}^{n}(\hat{y}_i - \overline{y})^2 + \sum_{i=1}^{n}(y_i - \hat{y}_i)^2$$

$$SSTotal \quad = \quad SSModel \quad + \quad SSE$$

Degrees of freedom:

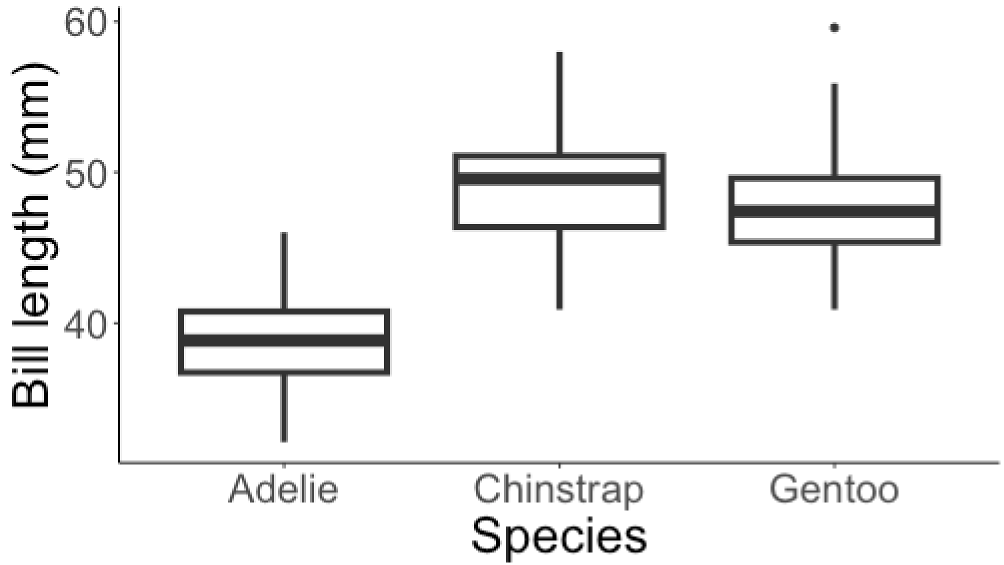
$$n - 1 \quad = \quad (p - 1) \quad + \quad (n - p)$$

($n =$ number of observations, $p =$ number of parameters in full model)

Test statistic:

$$F = \frac{\text{between group variance}}{\text{within group variance}} = \frac{SSModel/(p-1)}{SSE/(n-p)}$$

F test



Test statistic:

$$F = \frac{\text{between group variance}}{\text{within group variance}} = \frac{SSModel/(p-1)}{SSE/(n-p)} = 397.3$$

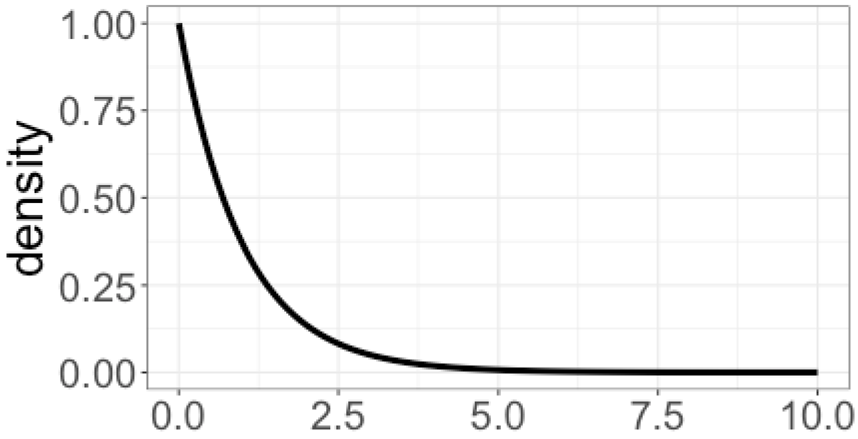
> How do I calculate a p-value?

Calculating p-values

If H_0 is true, $F = \dfrac{SSModel/(p-1)}{SSE/(n-p)} \sim F_{p-1,\, n-p}$

Degrees of freedom: $p - 1 = 2, n - p = 330$

$F_{2,\, 330}$ distribution:



p-value = $P(F_{2,\, 330} > 397.3) \approx 0$

Summary so far

Our F test compares two *nested* models:

Reduced model (H_0):

$$\text{bill length} = \beta_0 + \varepsilon$$

Full model (H_A):

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

Test statistic: $F = \dfrac{SSModel/(p-1)}{SSE/(n-p)} \sim F_{p-1,\, n-p}$

> What if I want to test other hypotheses?

Hypotheses for multiple regression

$$\begin{aligned} \text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} \\ + \beta_3 \text{body mass} + \varepsilon \end{aligned}$$

Research question: Is there any relationship between the predictors (species and body mass) and the response (bill length)?

> How do I turn this question into hypotheses about one or more model parameters?

Hypotheses for multiple regression

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$H_0 : \beta_1 = \beta_2 = \beta_3 = 0 \qquad H_A : \text{at least one of } \beta_1, \beta_2, \beta_3 \neq 0$

> What are the full and reduced models?

Full and reduced models

$H_0 : \beta_1 = \beta_2 = \beta_3 = 0 \qquad H_A : \text{at least one of } \beta_1, \beta_2, \beta_3 \neq 0$

Full model (H_A):

$$\begin{aligned} \text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} \\ + \beta_3 \text{body mass} + \varepsilon \end{aligned}$$

Reduced model (H_0):

$$\text{bill length} = \beta_0 + \varepsilon$$

> Note that the reduced model is *nested* inside the full model.

Hypotheses for multiple regression

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Research question: After accounting for species, is there a relationship between body mass and bill length?

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Research question: After accounting for species, is there a relationship between body mass and bill length?

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$H_0 : \beta_3 = 0 \qquad H_A : \beta_3 \neq 0$

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$H_0 : \beta_1, \beta_2 = 0 \qquad H_A : \text{at least one of } \beta_1, \beta_2 \neq 0$

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Reduced model (H_0):

$$\text{bill length} = \beta_0 + \beta_1 \text{body mass} + \varepsilon$$

> Note that the reduced model is *nested* inside the full model.

Hypotheses for multiple regression

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \beta_3 \text{body mass} + \varepsilon$$

Research question: Is the slope on body mass different for different species?

> Can I test this research question with this model?

Hypotheses for multiple regression

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} \\ + \beta_3 \text{body mass} + \varepsilon$$

Research question: Is the slope on body mass different for different species?

> Can I test this research question with this model?

No -- the model assumes that the slope is the same for each species!

> How can I change the model to test this research question?

Adding interaction terms

$$\begin{aligned}\text{bill length} = \beta_0 &+ \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} \\ &+ \beta_3 \text{body mass} + \beta_4 \text{body mass} \cdot \text{IsChinstrap} \\ &+ \beta_5 \text{body mass} \cdot \text{IsGentoo} + \varepsilon\end{aligned}$$

Research question: Is the slope on body mass different for different species?

> How do I turn this question into hypotheses about one or more model parameters?

Adding interaction terms

$$\begin{aligned}\text{bill length} = \beta_0 &+ \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} \\ &+ \beta_3 \text{body mass} + \beta_4 \text{body mass} \cdot \text{IsChinstrap} \\ &+ \beta_5 \text{body mass} \cdot \text{IsGentoo} + \varepsilon\end{aligned}$$

Research question: Is the slope on body mass different for different species?

> How do I turn this question into hypotheses about one or more model parameters?

$H_0 : \beta_4, \beta_5 = 0 \qquad H_A : \text{at least one of } \beta_4, \beta_5 \neq 0$

> What are the full and reduced models?

Testing interaction terms

$H_0 : \beta_4, \beta_5 = 0 \qquad H_A : \text{at least one of } \beta_4, \beta_5 \neq 0$

Full model (H_A):

$$\begin{aligned} \text{bill length} = \beta_0 &+ \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} \\ &+ \beta_3 \text{body mass} + \beta_4 \text{body mass} \cdot \text{IsChinstrap} \\ &+ \beta_5 \text{body mass} \cdot \text{IsGentoo} + \varepsilon \end{aligned}$$

Reduced model (H_0):

$$\begin{aligned} \text{bill length} = \beta_0 &+ \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} \\ &+ \beta_3 \text{body mass} + \varepsilon \end{aligned}$$

> Note that the reduced model is *nested* inside the full model.

Class activity

Work on the class activity (handout).

Class activity

Research question: Is the slope on Age the same for both boys and girls?

> What model should the researchers use to test this research question?

Class activity

Research question: Is the slope on Age the same for both boys and girls?

> What model should the researchers use to test this research question?

$$\text{Height} = \beta_0 + \beta_1 Age + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

> What are the full and reduced models that the researchers will compare?

Class activity

Full model:

$$\text{Height} = \beta_0 + \beta_1 Age + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

Reduced model:

$$\text{Height} = \beta_0 + \beta_1 Age + \beta_2 \text{IsMale} + \varepsilon$$

Class activity

$$\text{Height} = \beta_0 + \beta_1 Age + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

New research question: Is there *any* relationship between Age and Height, after accounting for gender?

> What are the full and reduced models that the researchers will compare?

Class activity

New research question: Is there *any* relationship between Age and Height, after accounting for gender?

Full model:

$$\text{Height} = \beta_0 + \beta_1 Age + \beta_2 \text{IsMale} + \beta_3 \text{Age} \cdot \text{IsMale} + \varepsilon$$

Reduced model:

$$\text{Height} = \beta_0 + \beta_1 \text{IsMale} + \varepsilon$$

What we've done so far

- Specifying hypotheses for multiple regression models
- Hypotheses imply full and reduced models

> How do we compare full and reduced models with a hypothesis test?

When our reduced model is intercept-only

We've already done this for one categorical predictor!

Reduced model (H_0):

$$\text{bill length} = \beta_0 + \varepsilon$$

Full model (H_A):

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

Test statistic: $F = \dfrac{SSModel/(p-1)}{SSE/(n-p)} \sim F_{p-1,\, n-p}$

> What if I want to compare different nested models?

Decomposing sums of squares

$$F = \frac{SSModel/(p-1)}{SSE/(n-p)}$$

- SSModel and SSE really refer to the *full* model ($\hat{y}_i$ are predictions from the *full* model, not the reduced model)
- So let's add subscripts to make that clear

$$\sum_{i=1}^{n}(y_i - \bar{y})^2 = \sum_{i=1}^{n}(\hat{y}_{i,full} - \bar{y})^2 + \sum_{i=1}^{n}(y_i - \hat{y}_{i,full})^2$$

$$SSTotal \quad = \quad SSModel_{full} \quad + \quad SSE_{full}$$

$$F = \frac{SSModel_{full}/(p-1)}{SSE_{full}/(n-p)} = \frac{(SSTotal - SSE_{full})/(p-1)}{SSE_{full}/(n-p)}$$

Decomposing sums of squares

$$F = \frac{(SSTotal - SSE_{full})/(p-1)}{SSE_{full}/(n-p)}$$

We can decompose sums of squares for the reduced model too:

$$\sum_{i=1}^{n}(y_i - \overline{y})^2 = \sum_{i=1}^{n}(\hat{y}_{i,reduced} - \overline{y})^2 + \sum_{i=1}^{n}(y_i - \hat{y}_{i,reduced})^2$$

$$SSTotal \quad = \quad SSModel_{reduced} \quad + \quad SSE_{reduced}$$

> For this test, the reduced model is
>
> $$\text{bill length} = \beta_0 + \varepsilon$$
>
> What are the predictions $\hat{y}_{i,reduced}$ for this intercept-only model?

Decomposing sums of squares

$$F = \frac{(SSTotal - SSE_{full})/(p-1)}{SSE_{full}/(n-p)}$$

We can decompose sums of squares for the reduced model too:

$$\sum_{i=1}^n (y_i - \overline{y})^2 = \sum_{i=1}^n (\hat{y}_{i,reduced} - \overline{y})^2 + \sum_{i=1}^n (y_i - \hat{y}_{i,reduced})^2$$

$$SSTotal \quad = \quad SSModel_{reduced} \quad + \quad SSE_{reduced}$$

Reduced model: $\text{bill length} = \beta_0 + \varepsilon$

Predictions: $\hat{y}_{i,reduced} = \overline{y}$

> If $\hat{y}_{i,reduced} = \overline{y}$, what are $SSModel_{reduced}$ and $SSE_{reduced}$?

Decomposing sums of squares

$$F = \frac{(SSTotal - SSE_{full})/(p-1)}{SSE_{full}/(n-p)}$$

We can decompose sums of squares for the reduced model too:

$$\sum_{i=1}^n (y_i - \overline{y})^2 = \sum_{i=1}^n (\hat{y}_{i,reduced} - \overline{y})^2 + \sum_{i=1}^n (y_i - \hat{y}_{i,reduced})^2$$

$$SSTotal \quad = \quad SSModel_{reduced} \quad + \quad SSE_{reduced}$$

Reduced model: $\text{bill length} = \beta_0 + \varepsilon$

Predictions: $\hat{y}_{i,reduced} = \overline{y}$

$$SSModel_{reduced} = 0 \qquad\qquad SSE_{reduced} = SSTotal$$

Decomposing sums of squares

$$\sum_{i=1}^{n}(y_i - \overline{y})^2 = \sum_{i=1}^{n}(\hat{y}_{i,reduced} - \overline{y})^2 + \sum_{i=1}^{n}(y_i - \hat{y}_{i,reduced})^2$$

$$SSTotal \quad = \quad SSModel_{reduced} \quad + \quad SSE_{reduced}$$

Reduced model: $\text{bill length} = \beta_0 + \varepsilon$

Predictions: $\hat{y}_{i,reduced} = \overline{y}$

$SSModel_{reduced} = 0 \qquad SSE_{reduced} = SSTotal$

$$\begin{aligned} F &= \frac{(SSTotal - SSE_{full})/(p-1)}{SSE_{full}/(n-p)} \\ &= \frac{(SSE_{reduced} - SSE_{full})/(p-1)}{SSE_{full}/(n-p)} \end{aligned}$$

Re-writing our F test

Our F test compares two *nested* models:

Reduced model (H_0):

$$\text{bill length} = \beta_0 + \varepsilon$$

Full model (H_A):

$$\text{bill length} = \beta_0 + \beta_1 \text{IsChinstrap} + \beta_2 \text{IsGentoo} + \varepsilon$$

Test statistic:

$$F = \frac{(SSE_{reduced} - SSE_{full})/(p-1)}{SSE_{full}/(n-p)} \sim F_{p-1,\, n-p}$$

- $p - 1 =$ difference in number of parameters between models
- $p =$ number of parameters in full model

Nested F test

To compare *any* two **nested** models:

Test statistic: $F = \dfrac{\frac{1}{\text{\# parameters tested}}(SSE_{reduced} - SSE_{full})}{\frac{1}{n-p} SSE_{full}}$

Distribution: $F_{\text{\# parameters tested}, \, n-p}$

- $n =$ number of observations in data
- $p =$ number of parameters in full model